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Unified engine for large-scale data analytics

GET STARTED

What is Apache Spark™?

Apache Spark™ is a multi-language engine for executing data engineering, data science, and machine learning on single-node machines or clusters.

Key features

Apache Spark™ - What is Spark | Apache Spark™ - Unified Engine

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APACHE Spark™

Apache Spark™

Apache Spark is a lightning-fast unified analytics engine for big data and machine learning. It was originally developed at UC Berkeley in 2009.

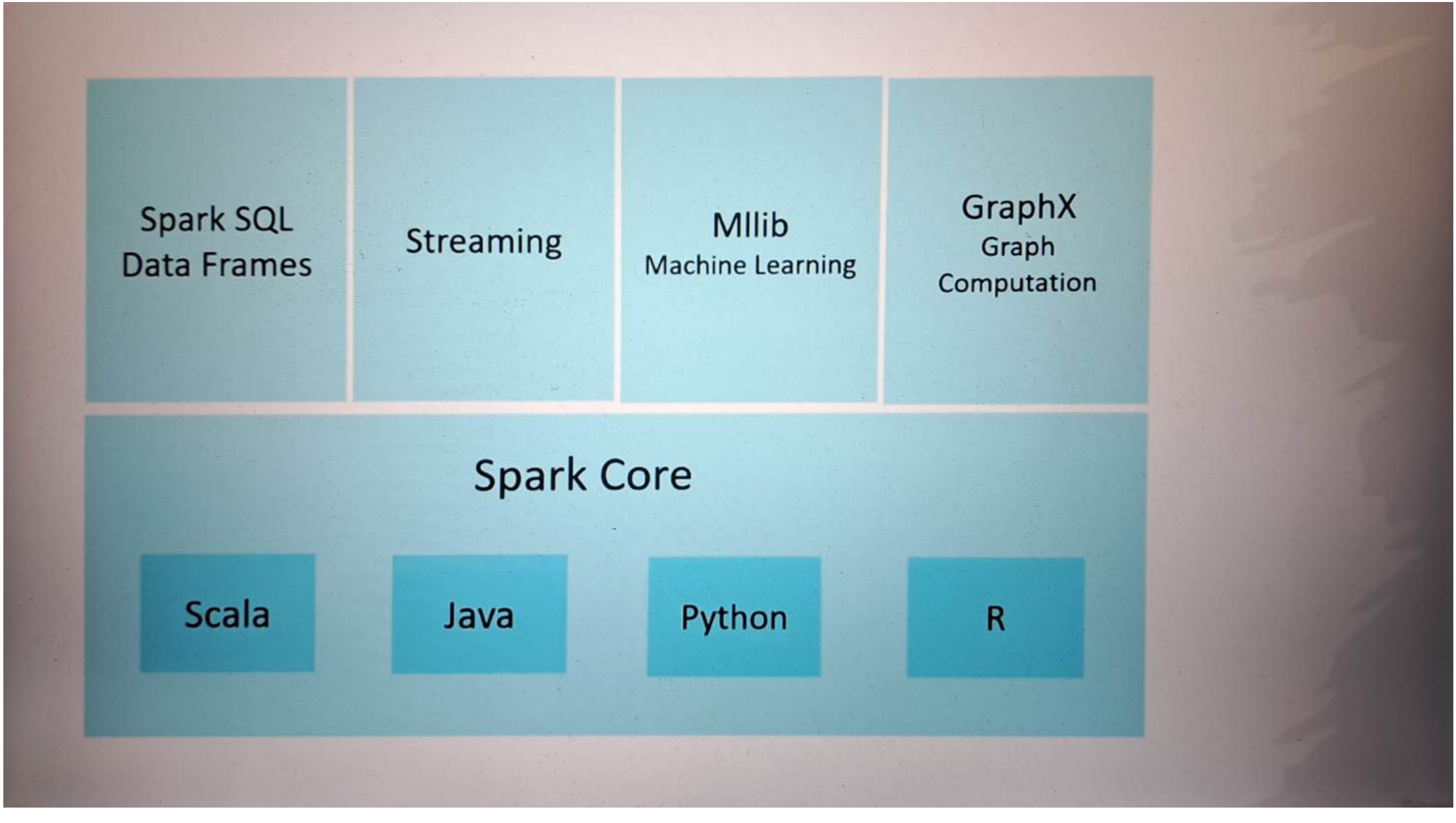
The largest open source project in data processing.

Since its release, **Apache Spark**, the unified analytics engine, has seen rapid adoption by enterprises across a wide range of industries. Internet powerhouses such as Netflix, Yahoo, and eBay have deployed Spark at massive scale, collectively processing multiple petabytes of data on clusters of over 8,000 nodes. It has quickly become the largest open source community in big data, with over 1000 contributors from 250+ organizations.

The team that started the Spark research project at UC Berkeley founded Databricks in 2013.

Apache Spark is 100% open source, hosted at the vendor-independent Apache Software Foundation. At Databricks, we are fully committed to maintaining this open development model. Together with the Spark community, Databricks continues to contribute heavily to the Apache Spark project, through both development and community evangelism.

What is Apache Spark?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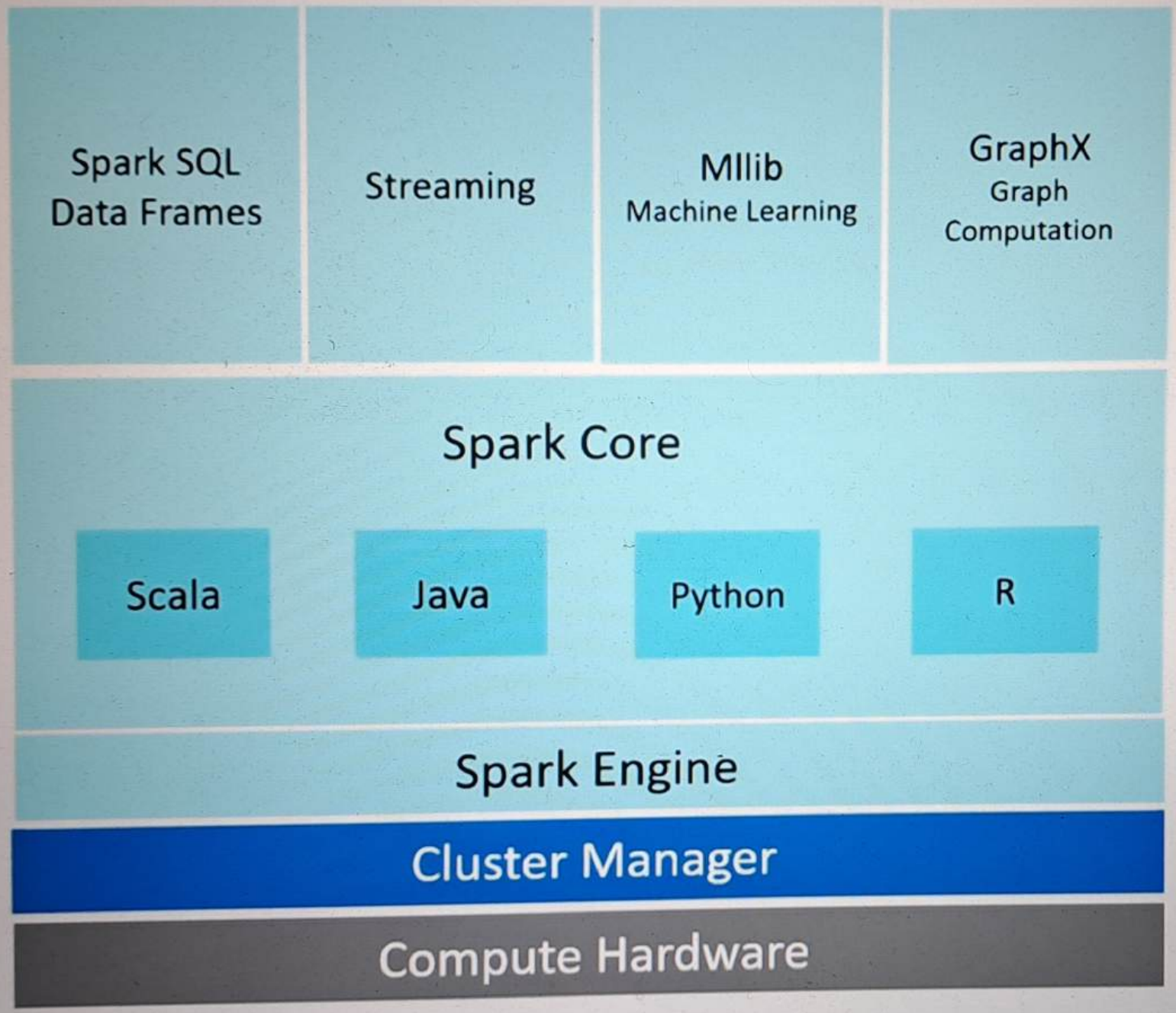
R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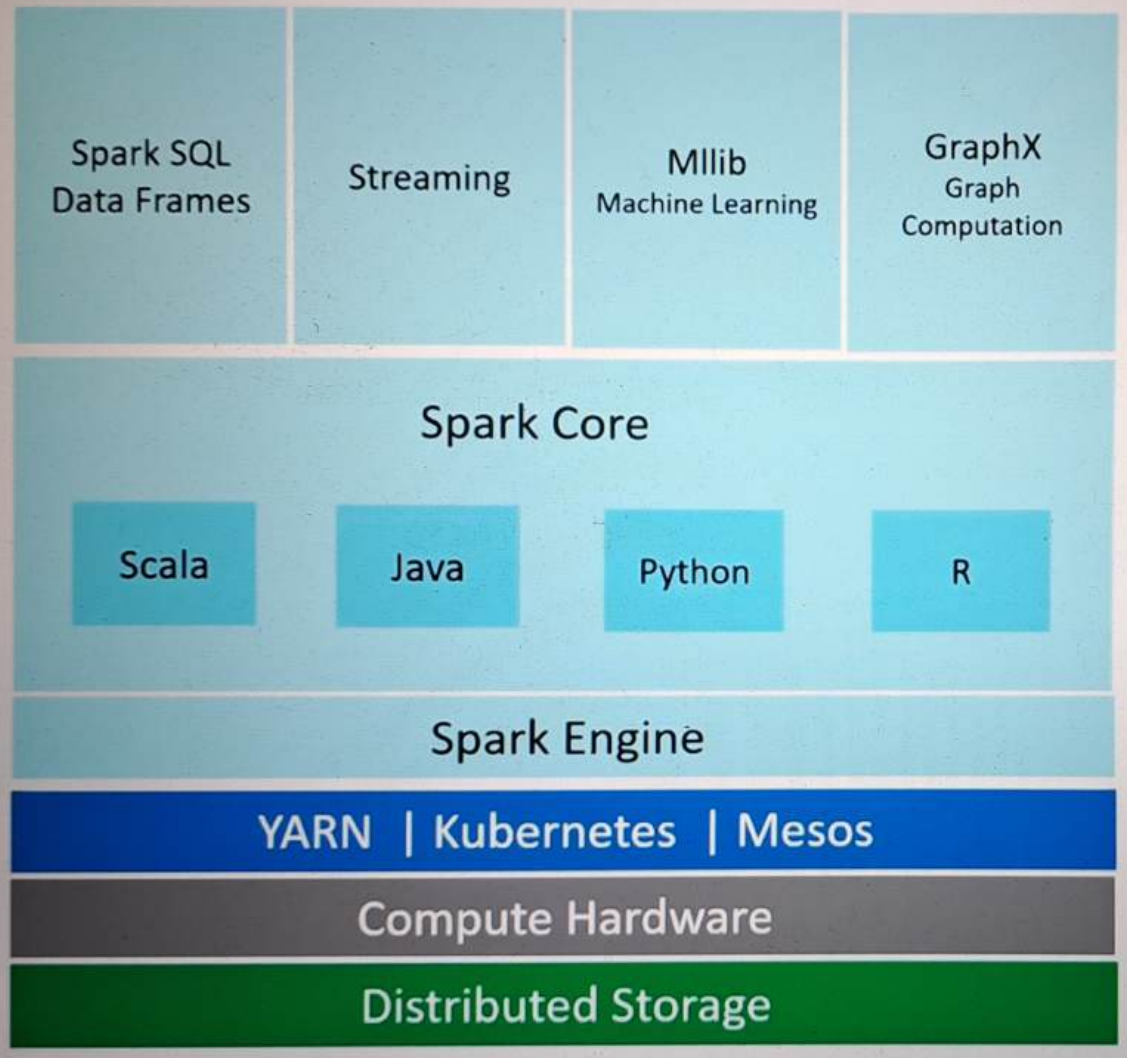
Compute Hardware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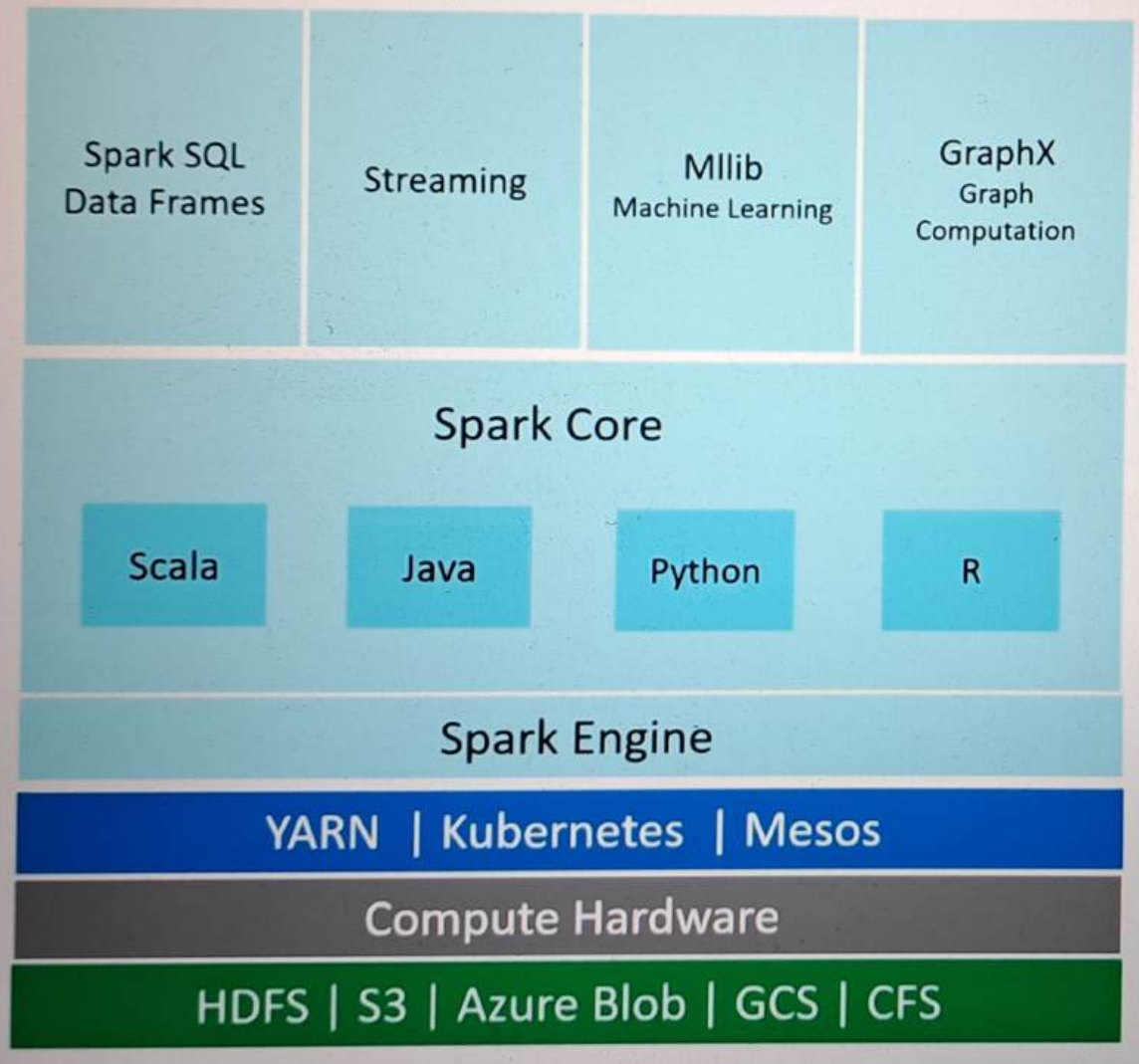
Compute Hardware
Distributed Storage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YARN | Kubernetes | Mesos
Compute Hardware
HDFS | S3 | Azure Blob | GCS | CFS

APACHE Spark™

1. Abstraction
2. Unified Platform
3. Ease of Use

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1. Spark on the Cloud Platform
2. Spark Cluster Management
3. Notebooks and Workspace
4. Administration Controls
5. Optimized Spark
6. Databases/Tables and Catalog
7. Databricks SQL Analytics
8. Delta Lake Integration
9. ML Flow
10. Industry vertical accelerators

- What do we do in the Spark application
- How does a typical Spark application code look
- How to write and execute your Spark application

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ScholarNest

What is Apache Spark?

Apache Spark™ is a multi-language engine for executing data engineering, data science, and machine learning on single-node machines or clusters.

Capabilities

- ANSI SQL
- Batch Processing API
- Stream Processing API
- Machine Learning API

Who is using Apache Spark?

Thousands of companies, including 80% of the Fortune 500, use Apache Spark™.

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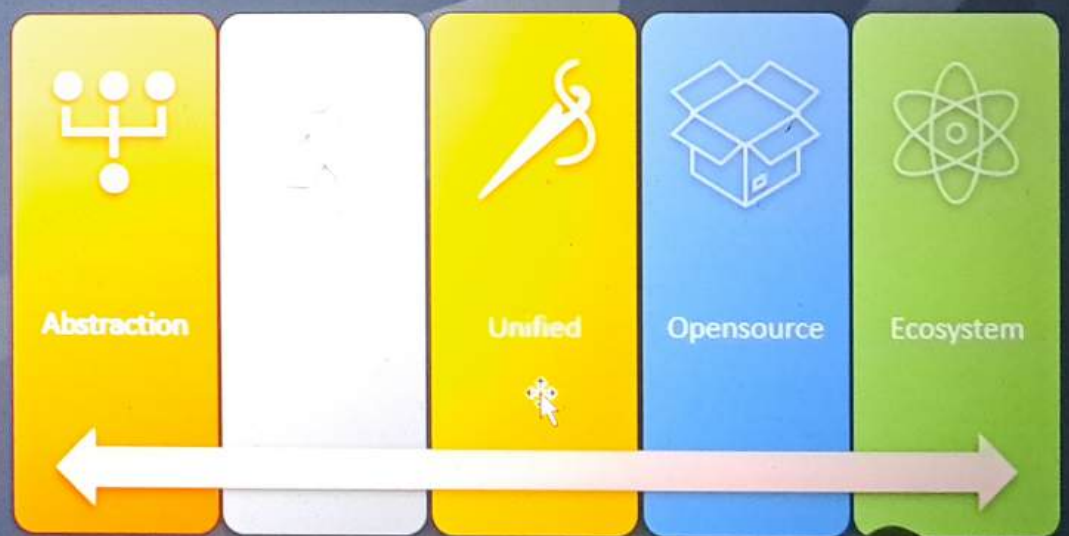
Why Apache Spark?



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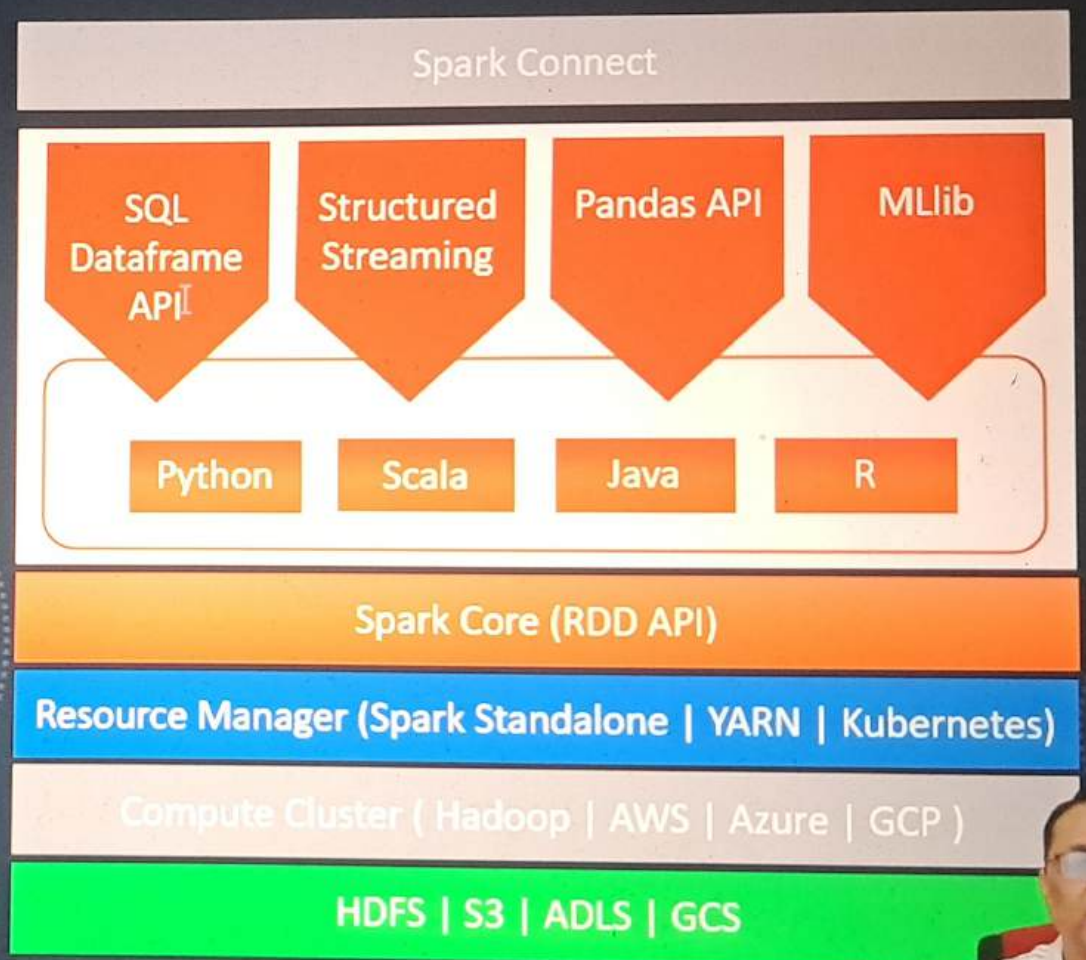
Spark Architecture

Unified Architecture



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What is Databricks?

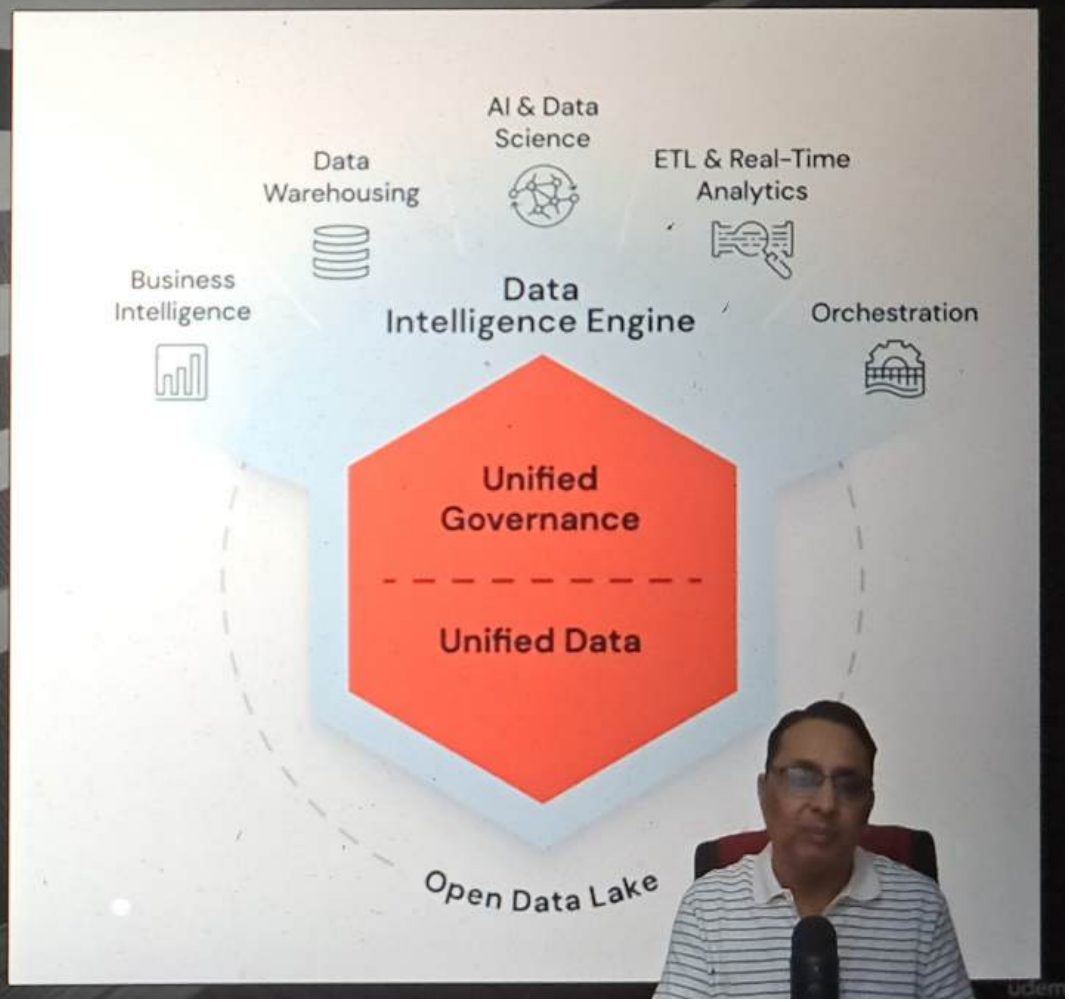
ScholarNest

Databricks Data Intelligence Platform is a unified, open analytics platform for building, deploying, sharing, and maintaining enterprise-grade data, analytics, and AI solutions at scale.

Managed open source integration

- Apache Spark
- ML Flow
- Delta Lake
- Unity Catalog
- Postgres

AWS
Azure
GCP



Databricks Platform Capabilities

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- Spark on the cloud platform
- On-demand cluster
- Delta lake Storage
- Workflows and ETL
- Data Warehouse
- ML and AI Capabilities
- Security and Governance
- Automation Tools
- Ecosystem and partners

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